

Technical Approach

Brainactiver UI/UX Redesign

Soft-glassmorphism reskin delivered as a CSS design-system rewrite - zero JSX changes

Project: sih-hackthon (Brainactiver) - cognitive training & brain-wellness platform
Date: 2026-09-18
Scope: src/index.css only (live stylesheet); no component/logic edits
Status: Implemented, build-verified, dev-server smoke-tested

1. Objective

The request was to **change the UI/UX design of the entire web application** and make it more attractive. After a codebase survey, the direction was agreed with the stakeholder as:

Soft glassmorphism (frosted-glass cards, soft gradients, calm senior-friendly feel), delivered as a **CSS design-system-only rewrite** - i.e. restyle every screen by rewriting the stylesheet, without touching any TSX component, state, or business logic.

Success criteria

- New cohesive look applied to **all screens** (auth, kiosk home, reminders, caregiver dashboard, progress, avatar editor, voice assistant, all 9 games).
- **Zero functional changes:** no JSX/TS edits, no behaviour or copy changes.
- Existing **theme engine** (5 themes) and **senior font-scaling** (3 sizes) keep working.
- `npm run build` passes; linter shows no new issues; dev server serves the app.

2. Discovery and baseline analysis

2.1 Technology stack (verified from package.json)

Layer	Technology
UI framework	React 19.2.8 + TypeScript (~6.0.2)
Build tool	Vite 8.2.2 (@vitejs/plugin-react)
Styling	Tailwind CSS v4 (@tailwindcss/vite) + one global stylesheet
Icons	lucide-react
Mobile shell	Capacitor 8 (Android)
Extras	jspdf, canvas-confetti
Lint	oxlint

2.2 Entry point and stylesheet graph (verified)

The production entry chain is: **index.html -> /src/main.tsx -> src/index.css + src/App.tsx**. A second copy of `index.css` (883 lines, truncated), `main.tsx` and `App.tsx` exists at the repository root, but nothing imports it, so it is a **stale duplicate outside the build graph**. Decision: modify only `src/index.css`; leave the root copies untouched.

2.3 Screen inventory (from the ActiveView union in src/App.tsx)

View key	Component	Main CSS hooks
language_select	auth/LanguageSelection	.login-container, .login-card, .language-option
login	auth/LoginPage	.role-option, .input-field, .btn-login, .demo-credential

kiosk	kiosk/ElderlyHome	.patient-hero-banner, .routine-card, .game-launch-card
reminders	reminders/ReminderManager	.routine-card, .btn-routine-action
progress	progress/PatientProgress	.progress-card, .activity-row
avatar_setup	avatar/AvatarEditor	.avatar-editor-card, .avatar-swatch
caregiver	caregiver/CaregiverDashboard	.dashboard-card, .csi-gauge-banner, .trend-bar
game_* (x9)	games/*Game	.game-arena-container, .btn-choice-option, .status-pill
(modal)	common/DisplaySettingsModal	.pin-modal-overlay, .pin-modal-box
(overlay)	voice/VoiceAssistant	.voice-fab, .voice-overlay-card, .voice-mic-orb

2.4 Theme and font-scaling engine

`src/App.tsx` writes `data-theme` (light / dark / comfort / forest / azure) and `data-font-scale` (normal / large / xlarge) onto ``; all visual variants are pure CSS overrides in the stylesheet, and settings persist through the `db` service. Note: the `highContrast` setting is stored but has **no CSS hook**, so it currently has no visual effect.

2.5 Styling contract

- All colours/surfaces come from **CSS custom properties** consumed both by the stylesheet and by **inline styles in TSX** (e.g. `var(primary-emerald)`, `var(emerald-surface)`, `var(muga-gold-bg)`, `var(bg-card)`, `var(text-secondary)`). Renaming any of them would silently break screens.
- Auth screens build tokens **dynamically**: `var(<color>-surface)` / `var(<color>-accent)` with color in {emerald, amber} (verified in LoginPage.tsx and UserTypeSelection.tsx). Both families must exist in every theme.

3. Technical approach and design decisions

3.1 Contract-preservation principle

Because the scope forbids JSX edits, the rewrite was constrained to **keep every existing CSS class selector and every CSS variable name**, changing only values and declarations. Before rewriting, all `className` literals were extracted from every `\*.tsx` file via regex and diffed against the selectors defined in `src/index.css`.

3.2 Pre-rewrite inventory result

Check	Result
Classes used in JSX but missing in CSS	1: .routine-card-icon (Heart icon in ElderlyHome)
Dynamic role tokens	Only emerald + amber - covered by existing variable families
Unused settings flag	highContrast stored, no CSS hook (documented, untouched)

The single missing class was **added** in the new system (gradient icon chip); everything else was restyled in place.

3.3 Glassmorphism adapted for senior users

- Translucency kept conservative (cards ~0.6-0.78 alpha) with strong text colours so readability is never sacrificed for effect.
- Existing accessibility features preserved: large touch targets, senior font-scale engine, semantic

correct/incorrect answer states.

- New a11y additions: a visible `:focus-visible` ring and a soft translucent scrollbar.

4. Implementation details

4.1 File changes

File	Change
src/index.css	Full rewrite: 1764 lines -> 2127 lines; all selectors/variables preserved; glass system added
src/*.tsx, index.html	No changes (per agreed scope)
index.css / main.tsx / App.tsx (repo root)	Deliberately untouched - stale duplicates outside the build graph

4.2 New design-token layer (added to :root, overridden per theme)

Token(s)	Purpose / value
--glass-blur	blur(22px) saturate(170%) - shared frosted-glass filter
--glass-border / --glass-border-soft	Translucent white edge lines for the glass look
--glass-highlight / --glass-strong	Sheen accents and stronger glass for nav/modals/login
--bg-card-solid	Opaque card colour for small chips that must stay fully legible
--aurora-a..d	Per-theme aurora blob colours driving the animated backdrop
--emerald / --amber	Role aliases derived via var() so they track the active theme automatically

4.3 Aurora backdrop

The page base is the parchment surface plus three fixed radial gradients on `body`. Two large animated blobs are painted with `::before`/`::after` on `.kiosk-container` and `.login-container` (fixed position, `z-index: -1`, `filter: blur(30px)`), drifting on a 20s/26s alternate `float-slow` keyframe. The containers use `isolation: isolate` with transparent backgrounds so the blobs sit above the page base but below all content, which is exactly what the frosted cards blur against.

4.4 Frosted surfaces

`backdrop-filter` (with `-webkit-` prefix for Safari/Capacitor WebViews) was applied to: the sticky navbar, routine/game/arena/dashboard/progress/avatar cards, login + modal + voice cards, nav/routine/choice/back buttons, status pills, badges, inputs and the PIN field.

4.5 Component highlights

- **Navbar**: translucent strong-glass bar with blur; brand title uses gradient text built from theme variables (teal -> blue -> gold) so it adapts to every theme.
- **Hero banner**: teal-to-blue gradient with two decorative radial sheens, white text with soft shadow, glass voice-help button.
- **Game cards**: glass + top-right sheen that scales up on hover; lift (-6px) with accent border and large soft shadow.
- **CTA buttons**: five gradient variants (emerald/amber/blue/purple/teal) with a gloss `::after` overlay, glow shadows, hover lift.
- **Game arena**: glass container with emerald-tinted border; choice options turn tinted glass on hover, with glowing correct / soft-red incorrect states.
- **Dashboard**: gradient CSI gauge banner; trend bars use vertical gradients with soft shadows.

- **Voice assistant:** gradient FAB with halo ring; pulsing mic orb; frosted overlay (`z-index 200`).
- **Motion:** kept `pulse-ring/slideUp/shake/spin/voice-pulse/fadeIn`; added `pop-in` for modals and `float-slow` for the backdrop.

4.6 Per-theme recalibration

Theme	Base / glass / text treatment
light (default)	Sky-tinted parchment; white glass; teal aurora; deep-teal text
dark	Deep-navy base; translucent navy glass; dimmed aurora; near-white text
comfort	Warm parchment; warm-white glass; amber/peach aurora; brown text
forest	Soft green base; white glass; green aurora; deep-green text
azure	Cool blue base; white glass; blue aurora; navy text

5. Verification and validation

Check	Command / method	Result
Type check + production build	npm run build (tsc -b && vite build)	PASS - CSS bundle 37.31 kB (gzip 7.07 kB)
Static analysis	npm run lint (oxlint)	No errors; only pre-existing hook warnings in games/App.tsx
Runtime smoke test	npm run dev + HTTP GET localhost:5173	HTTP 200, entry module present

6. Issues encountered and resolutions

Issue	Resolution
Build failed after rewrite: lightningcss 'Unexpected end of input' - a stray '</content>' line had been appended at end of src/index.css	Removed the stray line; verified EOF ends with '}' (2127 lines); rebuild green
Repo-root index.css/main.tsx/App.tsx look like live files but are stale duplicates	Confirmed via index.html entry (/src/main.tsx) and import graph; excluded from scope, left untouched
One JSX class (.routine-card-icon) had no styles	Added gradient icon-chip styling in the new system

7. Risks, limitations and next steps

Known limitations

- `backdrop-filter` over many cards can be GPU-heavy on low-end Android WebViews - monitor frame rates on the Capacitor build.
- Elements styled **inline in TSX** (page footer, navbar dropdown menus) inherit the new translucent `bg-card` but get no blur; they look glassy but flatter than class-styled cards. Unifying them needs a JSX-scope change.
- `highContrast` is persisted but has no CSS effect; `prefers-reduced-motion` is not honoured yet.

Recommended next steps

- 1. Visual QA pass per screen x per theme (light/dark/comfort/forest/azure) on desktop and a real Android device.
- 2. Decide the fate of the stale root-level `index.css` / `main.tsx` / `App.tsx` (delete or archive) to avoid

future confusion.

- 3. If approved, extend to JSX scope: footer/dropdown glass unification, reduced-motion support, high-contrast theme hook.
- 4. Optional: split the 2k-line stylesheet per route if it keeps growing, keeping the token layer central.